EXPERIMENT 4

Conduct task analysis for an app (e.g., online shopping) and document user flows. Create corresponding wireframes using Lucid chart

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Aim

The aim of this experiment is to perform task analysis for a Smart Library mobile application and design a single consolidated user flowchart that represents the major user activities including book search, reservation, renewal, and due-date checking.

The objective is to understand:

- How users interact with the system
- The step-by-step workflow
- Decision-making points
- System responses
- Overall usability structure

Procedure

The following procedure was followed to complete this experiment:

Step 1: Identify Application

Selected "Smart Library App" as the system for analysis.

Step 2: Define User Goals

Identified major tasks performed by students:

- Searching books
- Reserving books
- Renewing books

- Checking due dates

Step 3: Break Down Tasks

Each goal was divided into smaller step-by-step actions to understand system-user interaction clearly.

Step 4: Identify Decision Points

Critical decisions such as:

- Login validation
- Book availability
- Renewal eligibility
  were identified and marked using diamond symbols.

Step 5: Design User Flow

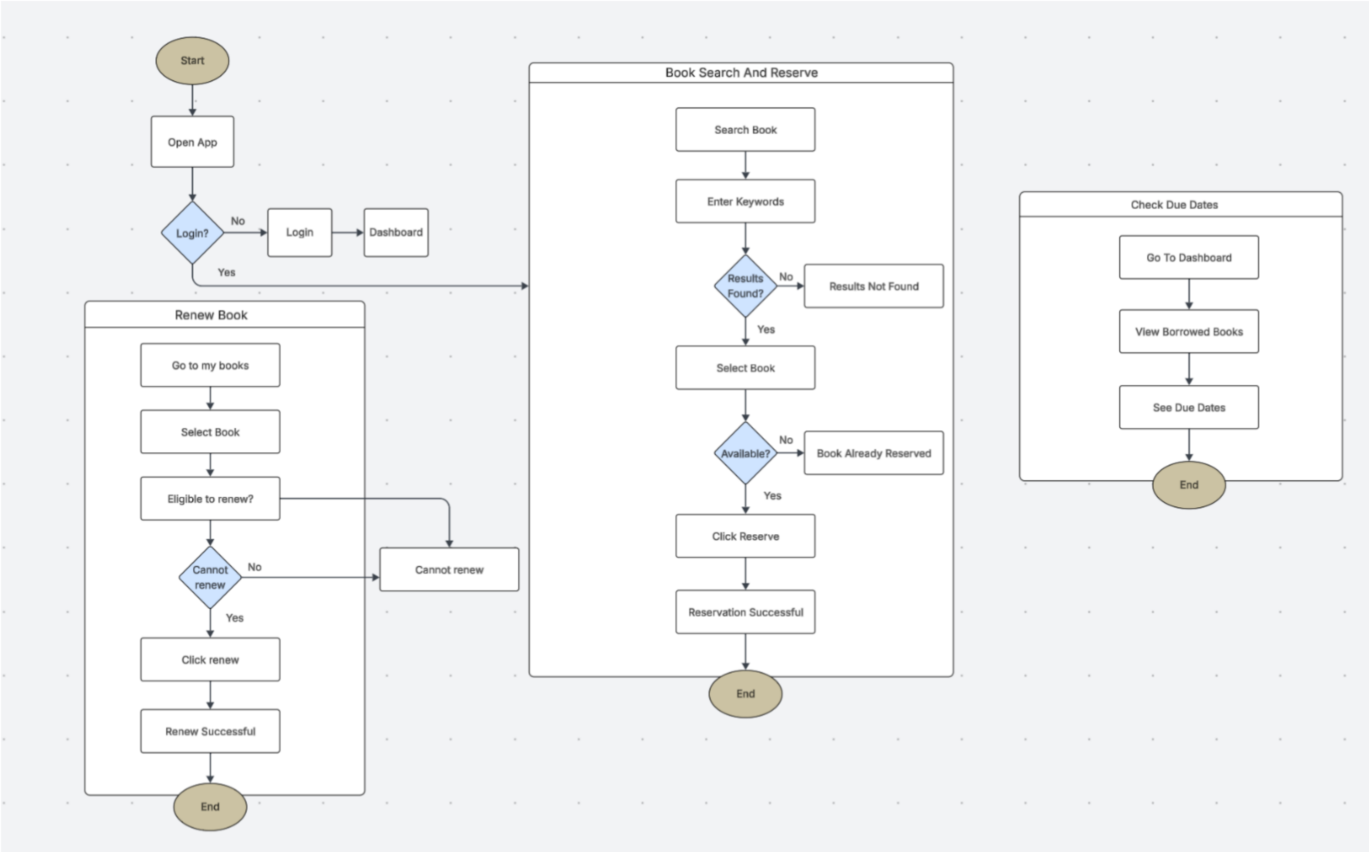
A single consolidated flowchart was created using:

- Oval → Start/End
- Rectangle → Process
- Diamond → Decision
- Arrows → Flow direction

Step 6: Tool Used

Lucid chart was used to create the final user flow diagram and low-fidelity structure.

User Flow for a Library System



Phase 1: Application Access

Step 1:

The process starts at Start.

Step 2:

User opens the application.

Step 3:

System checks whether the user is logged in.

Step 4:

If the user is NOT logged in:

- User enters login credentials.
- System verifies credentials.
- User is redirected to Dashboard.

Step 5:

If the user is already logged in:

- User directly proceeds to the main system functions.

Phase 2: Book Search and Reservation

Step 6:

User selects Search Book option.

Step 7:

User enters keywords (book name / author).

Step 8:

System checks whether results are found.

Step 9:

If results are NOT found:

- System displays “Results Not Found”.
- Process ends.

Step 10:

If results are found:

- User selects a book from the list.

Step 11:

System checks availability of the selected book.

Step 12:

If the book is NOT available:

- System shows “Book Already Reserved”.
- Process ends.

Step 13:

If the book is available:

- User clicks “Reserve”.

Step 14:

System confirms reservation.

Step 15:

Reservation Successful message is displayed.

Step 16:

Process ends.

Phase 3: Renew Book

Step 17:

User selects Go to My Books.

Step 18:

User selects the borrowed book.

Step 19:

System checks whether the book is eligible for renewal.

Step 20:

If the book is NOT eligible:

- System displays “Cannot Renew”.
- Process ends.

Step 21:

If eligible:

- User clicks “Renew”.

Step 22:

System updates the due date.

Step 23:

Renewal Successful message is displayed.

Step 24:

Process ends.

Phase 4: Check Due Dates

Step 25:

User goes to Dashboard.

Step 26:

User selects “View Borrowed Books”.

Step 27:

System displays borrowed books list.

Step 28:

User checks due dates.

Step 29:

Process ends.

Result:

The Smart Library App task analysis was completed successfully. A single flowchart was created to clearly represent the login, book search, reservation, renewal, and due-date checking processes.